mathematical laws can describe with exactitude the movements of the planets around the Sun, with a degree of accuracy even greater than any obtained in antiquity. We are now in 1600, and for the first time humanity finds out how to do something better than what was done in Alexandria more than a thousand years earlier.

While Kepler in the cold north calculates movements in the sky, in Italy it is with Galileo Galilei that the new science begins to take off. Exuberant, Italian, polemical, argumentative, highly cultured, exceptionally intelligent and overflowing with inventiveness, Galileo gets sent from Holland a new invention – the telescope – and makes a gesture which changes human history. He points it towards the sky.

Like Roy in *Blade Runner,* he sees things we people wouldn't believe: rings around Saturn, mountains on the moon, phases of Venus, moons orbiting Jupiter … Each of these phenomena renders Copernicus's ideas more plausible. Scientific instruments begin to open the myopic eyes of humankind on to a world vaster and more varied than it had as yet been able to conceive of.

But Galileo's grand idea is to make the logical deduction from the cosmic revolution brought about by Copernicus: convinced that the Earth is a planet like all others, Galileo reasons that, if movements in the heavens follow precise mathematical laws, and if the Earth is a planet like all others, and thus part of the heavens, then there must also exist precise mathematical laws governing the movements of objects *on Earth*.

Confident of the rationality of nature and of the Pythagorean-Platonic vision that nature is understandable through mathematics, Galileo decides to study *how* objects move on Earth when they are set free – that is, when they fall. Convinced that a relevant mathematical law must exist, he sets out to search for it, by trial and error. For the first time in the history of mankind, an *experiment* is made. Experimental science begins with Galileo. His experiment is simple: he lets objects fall; that is, he lets them follow what for Aristotle was their natural movement and seeks to measure precisely their falling speed.

The result is momentous: objects do not fall always at a constant speed, as everybody thought. Their speed, instead, gradually increases during the early part of the fall. In this phase, what is constant is not the speed of the fall but rather the *acceleration*, that is to say, the rate at which speed increases. And, magically, this acceleration turns out to be the same for all objects. Galileo completes a first rough measurement of this acceleration and finds it constant. Its value is approximately 9.8 metres per second per second, which is to say that for every second that an object falls, its speed increases by 9.8 metres per second. Keep in mind this number.